🏛️ 1. The Foundations of Probability

The Probability Space

All probability is defined on a 3-tuple $(\Omega, \mathcal{F}, P)$, called a **Probability Space**:

1. **Sample Space (Ω):** The set of *all possible individual outcomes* of an experiment.
   - *Example (Die Roll):* $\Omega = \{1, 2, 3, 4, 5, 6\}$
2. **Event Space ($\mathcal{F}$ or σ):** The set of all "events" we care about. An event is a subset of Ω.
   - *Example (Die Roll):* The event "roll an even number" is the set $\{2, 4, 6\}$.
3. **Probability Measure (P):** A function that assigns a probability (a number between 0 and 1) to every event in $\mathcal{F}$.

The Axioms of Probability (Kolmogorov Axioms)

For P to be a valid probability measure, it must follow three rules:

1. **Non-negativity:** For any event A, $P(A) \geq 0$.
2. **Normalization:** The probability of the entire sample space (a "certain event") is 1. $P(\Omega) = 1$.
3. **Countable Additivity:** For any sequence of *disjoint* events $A_1, A_2, \ldots$ (events that can't happen at the same time), the probability that *at least one* of them occurs is the sum of their individual probabilities.

$$P(A_1 \cup A_2 \cup \ldots) = P(A_1) + P(A_2) + \ldots$$

🎲 2. Random Variables (RVs)

A **Random Variable (RV)**, X, is a function that maps outcomes from the sample space (Ω) to real numbers ($\mathbb{R}$). It's a way of attaching a number to a random outcome.

- *Example (2 Coin Flips):* $\Omega = \{HH, HT, TH, TT\}$. Let X = "number of heads".
  - $X(HH) = 2$
  - $X(HT) = 1$
  - $X(TH) = 1$
  - $X(TT) = 0$

  The probability $P(X = 1)$ is the sum of probabilities of the events that map to it: $P(X = 1) = P(HT) + P(TH)$.

Discrete RVs

A discrete RV has a countable number of possible values (e.g., 0, 1, 2, 3...).

- **Probability Mass Function (PMF):** $p(x) = P(X = x)$. This gives the *exact probability* of a specific value.
- **Cumulative Distribution Function (CDF):** $F(x) = P(X \leq x)$. This is the "accumulated" probability. For a discrete RV, it's a **step function** that jumps at each possible value.

Continuous RVs

A continuous RV has an uncountable number of possible values in a range (e.g., height, temperature).

- **Probability Density Function (PDF):** $f(x)$.
  - This is **NOT a probability**. It represents **likelihood** or "probability density". It can be greater than 1.
  - $P(X = x) = 0$ for any *exact* value x.
  - To get a probability, you must **integrate** over a range: $P(a \leq X \leq b) = \int_a^b f(x)\,dx$.
- **Cumulative Distribution Function (CDF):** $F(x) = P(X \leq x)$.
  - $F(x)$ is a continuous, non-decreasing function from 0 to 1.
  - The PDF is the derivative of the CDF: $f(x) = \frac{d}{dx}F(x)$.

📈 3. Properties of RVs: Expectation & Variance

- **Expectation (Mean, $E[X]$, μ):** The long-term average value of the RV; the "center of mass" of its distribution.
  - **Discrete:** $E[X] = \sum_x x \cdot P(X = x)$
  - **Continuous:** $E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\,dx$
- **Law of the Unconscious Statistician (LOTUS):** To find the expectation of a *function* of an RV, $g(X)$, you don't need to find $g(X)$'s distribution.
  - **Discrete:** $E[g(X)] = \sum_x g(x) \cdot P(X = x)$
  - **Continuous:** $E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f(x)\,dx$
- **Variance ($\mathrm{Var}(X)$, σ^2):** Measures the "spread" of the distribution. It's the expected squared distance from the mean.
  - **Definition:** $\mathrm{Var}(X) = E[(X - \mu)^2]$
  - **Computational Formula:** $\mathrm{Var}(X) = E[X^2] - (E[X])^2$

📊 4. Common Probability Distributions

Discrete Distributions

- **Bernoulli(p):** A single trial with two outcomes (Success=1, Failure=0).
  - $P(X = 1) = p$, $P(X = 0) = 1 - p$.
  - $E[X] = p$, $\mathrm{Var}(X) = p(1 - p)$.
- **Binomial(n, p):** The number of successes in n independent Bernoulli trials.
  - **PMF:** $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
  - $E[X] = np$, $\mathrm{Var}(X) = np(1 - p)$.
- **Poisson(λ):** The number of events in a fixed time/space interval, given an average rate λ.
  - **PMF:** $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$
  - $E[X] = \lambda$, $\mathrm{Var}(X) = \lambda$.
- **Geometric(p):** The number of trials *until* the first success.
  - **PMF:** $P(X = k) = (1 - p)^{k-1} p$
  - $E[X] = 1/p$.
- **Hypergeometric:** Sampling n items *without replacement* from a population of N items containing K successes. Models the number of successes k in the sample.

Continuous Distributions

- **Uniform(a, b):** All values in the range $[a, b]$ are equally likely.
  - **PDF:** $f(x) = \frac{1}{b-a}$ for x in $[a, b]$.
  - $E[X] = \frac{a+b}{2}$.
- **Exponential(λ):** The waiting time *until* the next event in a Poisson process.
  - **PDF:** $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.
  - $E[X] = 1/\lambda$.
  - **Memoryless Property:** $P(X > s + t \mid X > s) = P(X > t)$. The fact that you've already waited s seconds doesn't change the probability of waiting t more.
- **Normal ($N(\mu, \sigma^2)$):** The "bell curve". Defined by its mean μ and variance σ^2.
  - **Standard Normal ($N(0, 1)$):** A special case with $\mu = 0$ and $\sigma^2 = 1$. We use Z for this.

🤝 5. Multiple Random Variables

- **Joint Distribution:** Describes the behavior of two or more RVs.
  - **Joint PDF:** $f_{X,Y}(x, y)$. Probability is the *volume* under this surface.
- **Independence:** X and Y are independent if knowing one tells you nothing about the other.
  - **Formal Test:** $f_{X,Y}(x, y) = f_X(x) f_Y(y)$. The joint PDF factors into the marginal PDFs.
- **Covariance:** Measures the *linear* relationship between X and Y.
  - **Definition:** $\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$
  - **Computational Formula:** $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$
  - If $\mathrm{Cov}(X, Y) > 0$, they tend to move together.

- If $\mathrm{Cov}(X, Y) < 0$, they tend to move in opposite directions.
- **Independence $\implies$ Covariance = 0.**
- **Covariance = 0 $\not\Longrightarrow$ Independence.** (Could be a non-linear relationship, e.g., $Y = X^2$).

- **Variance of a Sum:** $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$.
  - If X and Y are **independent**, $\mathrm{Cov}(X, Y) = 0$, so $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.
- **Order Statistics:** The distribution of the i-th smallest RV, $X_{(i)}$, from a sample $X_1, \ldots, X_n$.
  - **PDF:** $f_{X_{(i)}}(x) = \frac{n!}{(i-1)!(n-i)!}[F(x)]^{i-1}[1-F(x)]^{n-i}f(x)$

🧬 6. Generating Functions & Derived Distributions

Moment Generating Function (MGF)

A unique "fingerprint" for a distribution, defined as $\phi(t) = E[e^{tX}]$.

1. **Generates Moments:** The k-th derivative evaluated at $t = 0$ gives the k-th moment.
   - $\phi'(0) = E[X]$
   - $\phi''(0) = E[X^2]$
   - $\mathrm{Var}(X) = \phi''(0) - [\phi'(0)]^2$
2. **Sums of Independent RVs:** The MGF of a sum is the *product* of the MGFs.
   - $\phi_{X+Y}(t) = \phi_X(t) \cdot \phi_Y(t)$
3. **Uniqueness:** If two RVs have the same MGF, they *must* have the same distribution.

Distribution	MGF $\phi(t)$
Binomial(n, p)	$(pe^t + 1 - p)^n$
Poisson(λ)	$e^{\lambda(e^t - 1)}$
Exponential(λ)	$\frac{\lambda}{\lambda - t}$
Normal(μ, σ^2)	$e^{\mu t + \sigma^2 t^2/2}$

- **Laplace Transform:** $g(t) = E[e^{-tX}]$. This is just $\phi(-t)$ and is guaranteed to exist for non-negative RVs.

Derived Distributions

- **Chi-Squared (χ_n^2):** The distribution of the sum of n independent, *squared* Standard Normal ($N(0,1)$) variables.
  - $Y = Z_1^2 + Z_2^2 + \ldots + Z_n^2 \sim \chi_n^2$
  - **MGF:** $\phi_Y(t) = (1-2t)^{-n/2}$
- **Cochran's Theorem:** A key theorem for statistics. If you take a sample $X_1, \ldots, X_n$ from a $N(\mu, \sigma^2)$ population:
  i. The sample mean $\bar{X}$ and sample variance S^2 are **independent**.
  ii. The scaled sample variance $\frac{(n-1)S^2}{\sigma^2}$ follows a χ_{n-1}^2 distribution (with $n-1$ degrees of freedom).

🏔️ 7. Limit Theorems

These theorems describe the behavior of RVs as the sample size $n \to \infty$.

The Inequalities

These provide "bounds" on probabilities when we have limited information.

- **Markov's Inequality:** (Requires only the mean)
  For a **non-negative** RV X, $P(X \geq a) \leq \frac{E[X]}{a}$.
- **Chebyshev's Inequality:** (Requires mean and variance)
  Gives a bound on the probability of being "far" from the mean.

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

The Laws of Large Numbers (LLN)

These theorems state that the **sample mean $\bar{X}$ converges to the true population mean** μ as n gets large. This is the mathematical reason *why* we can use experiments to estimate real-world probabilities.

- **Weak Law (WLLN):** (Convergence in Probability) The probability of the sample mean being "far" from the true mean gets smaller and smaller.

$$P(|\bar{X}_n - \mu| \geq \epsilon) \to 0 \text{ as } n \to \infty$$

- **Strong Law (SLLN):** (Convergence with Probability 1) The probability that the sample mean *doesn't* converge to the true mean is 0. It's a stronger guarantee that the sample mean will eventually get close and *stay* close.

$$P(\lim_{n \to \infty} \bar{X}_n = \mu) = 1$$

The Central Limit Theorem (CLT)

The "crown jewel" of probability. It states that the **sum** (or **average**) of a large number n of independent and identically distributed (i.i.d.) RVs will be **approximately Normally distributed**, *regardless* of the original distribution.

- This is why the bell curve is so common in nature.
- **De Moivre-Laplace Theorem:** A special case of the CLT. It says that for large n, a **Binomial(n, p)** distribution can be approximated by a **Normal** distribution with $\mu = np$ and $\sigma^2 = np(1-p)$.